STAT/MA 41600

In-Class Problem Set #35: November 9, 2026

1. While testing a new beverage for its sugar content, the amount of sugar (per can of soda) has a Normal distribution with mean 39 grams and standard deviation 2 grams.

1a. If a person purchases 5 cans of soda, and the amount of sugar is independent between the cans of soda, what is the probability that all 5 cans of soda contain less than 40 grams of sugar each?

1b. If a person (instead) purchases cans of soda until they find the first can of soda that has 40 grams of sugar or more, how many cans of soda do they expect to need to purchase? (You may assume that the amount of sugar is independent between the cans of soda.)

2. A veterinarian considers cats to be fat if they weigh more than 15 pounds. Suppose that the weights of cats are considered to be independent and Normally distributed, each with average weight of 12 pounds and standard deviation of 5 pounds. If 10 people bring their cats to the veterinarian on a given day, what is the distribution of the number of fat cats? What are the parameters of that random variable?

3a. In the situation from Question 1, find a constant "c" so that the probability of a can of soda having between $39 - c$ grams of sugar and $39 + c$ grams of sugar is 25%.

3b. In the situation from Question 2, find a constant "c" so that the probability of a cat weighing between $12 - c$ pounds and $12 + c$ pounds is 92%.

4. Suppose that the heights of blades of grass are Normally distributed, with each height having expected value 4 inches and standard deviation 0.75 inches.

If 100 blades of grass are collected and their heights are assumed to be independent, what is the probability that all 100 of them have height 2 inches or more?

5a. In the situation from Question 1, find the probability that a can of soda has more than 42.6 grams of sugar, given that it has more than 40.8 grams of sugar.

5b. In the situation from Question 2, find the probability that a cat weighs more than 14 pounds, given that it weighs more than 13 pounds.

5c. In the situation from Question 4, find the probability that a blade of grass is shorter than 3.2 inches, given that it is shorter than 3.8 inches.